

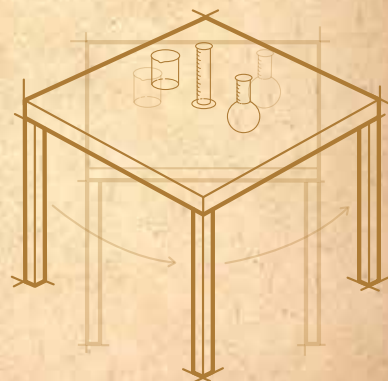


The Universe According to Emmy Noether

How a deceptively simple theorem from a century ago still shapes modern physics.

BY **STEVE NADIS**

ILLUSTRATIONS BY **JAY SMITH**





In her 53 years, many spent bucking a system that impeded her pursuit of mathematics, Noether had an extraordinary impact on both algebra and physics. There's no telling what else she might have accomplished if society and fate had been more kind.

In 1915, two of the world's top mathematicians, David Hilbert and Felix Klein, invited Emmy Noether to the University of Göttingen to investigate a puzzle. A problem had cropped up in Albert Einstein's new theory of gravity, general relativity, which had been unveiled earlier in the year. It seemed that the theory did not adhere to a well-established physical principle known as conservation of energy, which states that energy can change forms but can never be destroyed. Total energy is supposed to remain constant. Noether, a young mathematician with no formal academic appointment, gladly accepted the challenge.

She resolved the issue head-on, showing that energy may not be conserved "locally" — that is, in an arbitrarily small patch of space — but everything works out when the space is sufficiently large. That was one of two theorems she proved that year in Göttingen, Germany. The other theorem, which would ultimately have a far greater impact, uncovered an intimate link between conservation laws (such as the conservation of energy) and the symmetries of nature, a connection that physicists have exploited ever since. Today, our current grasp of the physical world, from subatomic particles to black holes, draws heavily upon this theorem, now known simply as Noether's theorem.

"It is hard to overstate the importance of Noether's work in modern physics," Durham University physicist Ruth Gregory said a century later. "Her basic insights on symmetry underlie our methods, our theories and our intuition. The link between symmetry and conservation is how we describe our world."

A LIFE OF WORK

Who was this woman, called upon by two renowned mathematicians to help rescue Einstein's masterwork? On the face of it, Noether (pronounced NUR-tuh) appears to have been a curious choice. She did not have an actual job in mathematics and was barely able to get an education in the field. Yet she had published some important papers, and Hilbert felt that her expertise could help clear up the problem with general relativity.

Born in Erlangen, Germany, in 1882, Noether hoped to follow in the footsteps of her mathematician father, Max. But German universities did not admit women when she reached college age, so Noether had to audit classes instead. Eventually she did so well in the final exams that she earned an undergraduate degree.

In 1904 she was permitted to enroll in a doctoral program at the University of Erlangen. She received a Ph.D. in 1907 and spent nearly eight years working there without pay or an official position, relying on her family for financial support while occasionally filling in for her father as a substitute teacher. After her trip to Göttingen in 1915, she stayed on as a lecturer, again receiving no pay.

After years of working essentially as a volunteer, Noether finally became an untenured associate math professor in 1922 at Göttingen, where she was allotted a modest salary. But 11 years later, she lost her job when she and other Jews were cast out of academia in Nazi Germany. Soon after, she left the country and landed a job at Bryn Mawr College in Pennsylvania, with the help of Einstein. She died just 18 months later due to complications from surgery to remove an ovarian cyst.

In her 53 years, many spent bucking a system that impeded her pursuit of mathematics, Noether had an extraordinary impact on both algebra (her main field) and physics. There's no telling what else she might have accomplished if society and fate had been more kind. Nevertheless, her body of work was more than enough to secure her place in the pantheon of great scientists, with her namesake theorem perhaps her most durable contribution.

THE THICK OF THE THEOREM

Noether's theorem is a simple and elegant link between seemingly unrelated concepts that is, today, almost obvious to physicists. But nonphysicists can get the gist of it, too.

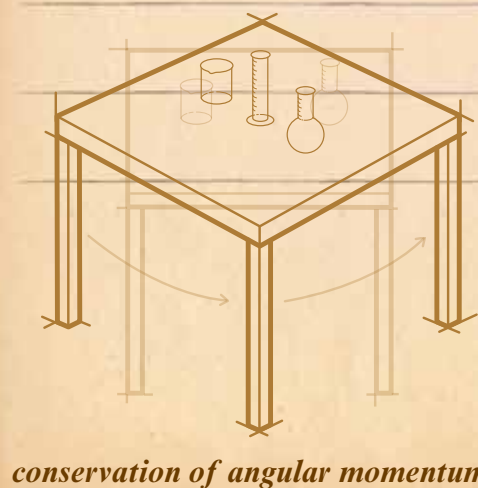
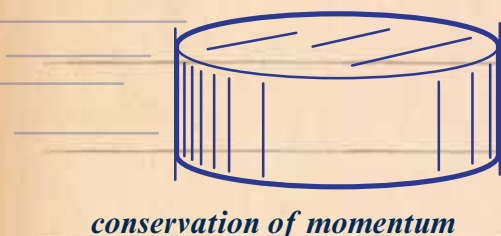
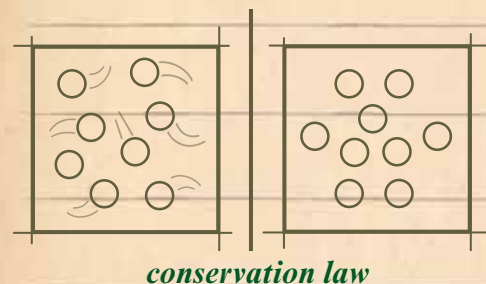
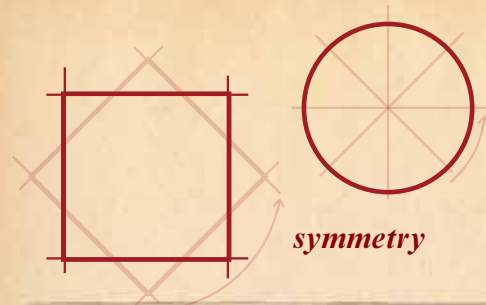
Basically, it states that every "continuous" symmetry in nature has a corresponding conservation law, and vice versa. Let's break down a few of those terms. **Symmetry**, in this context, refers to an operation that can be done to an object or system that leaves it unchanged. Rotating a square by 90 degrees is an example of "discrete" symmetry. The square still looks the same, whereas a 45-degree rotation yields something different (commonly called a diamond). A circle, on the other hand, possesses continuous symmetry since rotating it by any degree, or a fraction thereof, doesn't alter its appearance. This is the kind of symmetry to which Noether's theorem applies. A **conservation law**, meanwhile, refers to a physical quantity that remains fixed and hence does not fluctuate over time. Energy, for example, cannot be created or destroyed; once you've computed its value, there's no need to repeat the calculation.

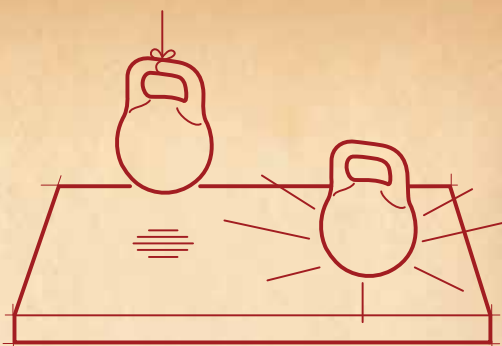
Noether's theorem uncovered a hidden relationship between two basic concepts — symmetries and conserved quantities — that until then had been treated separately. The theorem provides an explicit mathematical formula for finding the symmetry that underlies a given conservation law and, conversely, finding the conservation law that corresponds to a given symmetry.

Here's a glimpse of the theorem in action: Imagine a hockey puck gliding along a perfectly smooth, endless and frictionless sheet of ice. Let's further suppose that no external forces are acting on the puck whatsoever. Under these idealized conditions, the puck will continue to glide in a straight line without ever slowing down. Its momentum, the product of its mass and velocity, will be retained, or conserved. The only thing that could cause the puck to alter its course, or to gain or lose speed, would be if space itself — the surface of the ice, in this case — were to vary. Nothing will change, however, if the ice remains smooth and space remains unchanged.

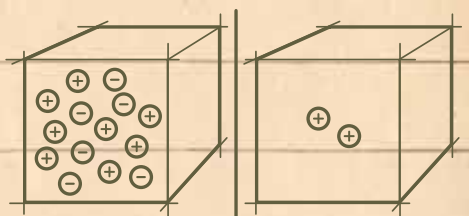
Noether's theorem shows that the puck's **conservation of momentum** is tied to its "symmetry of space translation," which is another way of saying that physics is not affected by linear movements (or translations) within a uniform space. The puck moves the same way on one part of the smooth ice as on another.

Similarly, Noether's theorem shows that symmetry under rotation, or rotational invariance, leads to the **conservation of angular momentum**, which measures how much an object is rotating. Physics, in other words, has no preferred direction. If you do an experiment on a table and then rotate that table by 45 degrees, or indeed by any amount, the experimental results will not differ. The theorem also links the symmetry of "time translation" to the

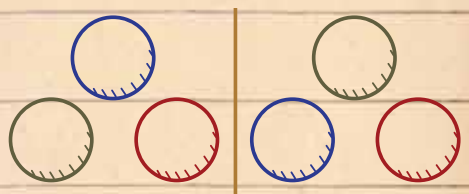




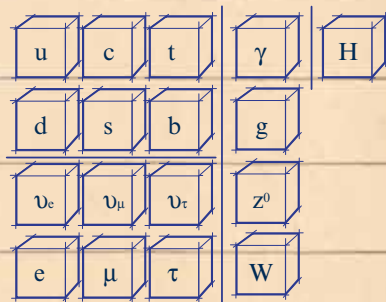
conservation of energy



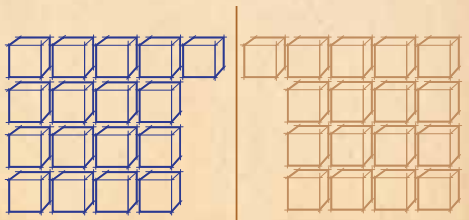
conservation of electric charge



color symmetry



Standard Model



supersymmetry

conservation of energy, so physics also doesn't care if you do an experiment today, next Tuesday or the third Sunday in October.

Physicists had known about the conservation of momentum, angular momentum and energy long before Noether's theorem came along. They are foundational precepts of classical mechanics. But it was not known that these hallowed laws shared a common origin, each bound to a particular symmetry. This new insight, which sprang from Noether's work, is a guiding principle that permeates physics research, while informing our views of the universe at large.

PUTTING IT ALL TOGETHER

Noether's theorem applies not only to these intuitive symmetries — rotations and shifts in time or space — but also to more abstract, "internal" symmetries that underlie the forces of nature.

For example, the **conservation of electric charge**, a central tenet of the theory of electromagnetism, stems from a symmetry related to details of the particle's spin. Another example: A symmetry called isospin that allows electrons to be substituted for neutrinos, and neutrinos for electrons, helped physicists develop a theory in the 1960s that unified the electromagnetic force and the weak force (which explains particle decays and radioactive processes) into a single electroweak force. The conserved quantity here is "hypercharge" — a kind of charge, analogous to electric charge, that is associated with this electroweak force. A decade later, physicists devised a theory for the strong nuclear force, which binds protons and neutrons in the atomic nucleus. At the heart of this force is something called **color symmetry**. (Color is a property of the quarks that make up protons and neutrons, which physicists view as another kind of charge.)

In the 1970s, physicists put all the known particles (including a few whose existence had not yet been confirmed, like the Higgs boson) and the forces that govern their interactions — the electromagnetic, weak and strong — into a single theoretical framework known as the **Standard Model**.

According to Stanford University physicist Michael Peskin, Noether's theorem was a basic tool in the construction of this amazingly successful model. "In quantum mechanics, you identify two or three particles that are supposed to be tied by a symmetry and then see if the inferred conservation law is valid. That's how you learn whether it is a real symmetry of nature, and that's how the Standard Model was built" — through a cumulative, step-by-step process like this. It's also how researchers are now trying to move forward.

A SUPER LEGACY

The hunt is on to find new particles and deeper, broader symmetries from which they stem, a process in which Noether's theorem continues to play a pivotal role. Much of the current effort focuses on looking for signs of **supersymmetry** — a theory that postulates a symmetry between the particles that make up matter (fermions) and the particles that transmit forces, like electromagnetism (bosons). If supersymmetry is right, every known fermion has a yet-to-be-observed bosonic "superpartner," and every known boson, likewise, has an as-yet-unseen fermionic superpartner.

The hypothetical supersymmetric particles, which physicists hope to discover at giant particle accelerators like the Large Hadron

Collider, would be “a reflection of all the Standard Model particles, using a mirror that is slightly distorted,” explains Joseph Incandela, a physicist at the University of California, Santa Barbara. “The particles on the other side of the mirror look just like Standard Model particles, except that their spins have been slightly shifted.”

One possibility that has been associated with this assumed symmetry, says Incandela, is the conservation of something called *r* parity, which implies that the lightest supersymmetric particle has to be stable and can never decay. If *r* parity is indeed conserved, every ordinary particle’s unseen supersymmetric partner will eventually decay into the lightest supersymmetric particle, which sticks around forever. That particle, whatever it may be, would be available in abundant quantities and could thus be a good candidate for the mysterious dark matter believed to account for more than one-quarter of the stuff in the universe.

ILLUMINATING BLACK HOLES

Noether’s theorem, however, is crucial to more than just the search for new particles; it extends to all branches of physics. Harvard physicist Andrew Strominger, for example, has identified an infinite number of symmetries related to **soft particles**, which are particles that have no energy. These particles come in two varieties: soft photons (particles that transmit the electromagnetic force) and soft gravitons (particles that transmit the gravitational force). Recent papers by Strominger and his colleagues, Stephen Hawking and Malcolm Perry of Cambridge University, suggest that material falling into a black hole adds soft particles to the black hole’s boundary, or **event horizon**. These particles would in effect serve as recording devices that store information, providing clues about the original material that went into the black hole.

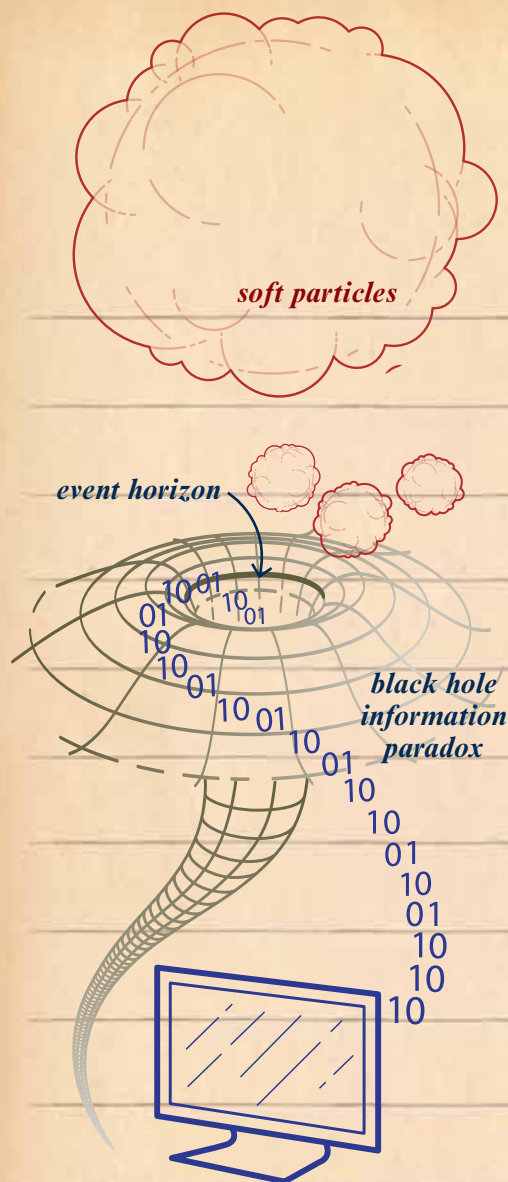
The idea proposed by the three physicists offers a new strategy for addressing a long-standing conundrum in physics known as the **black hole information paradox**. Hawking showed in the 1970s that every black hole will eventually evaporate and disappear, potentially destroying all the information the object once contained about how it formed and evolved over time. The permanent loss of information in Hawking’s scenario was troubling to theorists — including Hawking — as it would violate a cherished law of quantum physics holding that information, like energy, is always conserved.

The presence of soft particles along the event horizon, and their attendant symmetries, may point toward a way out of this dilemma. “We quickly realized through Noether’s theorem that there were conservation laws corresponding to the new symmetries that place very stringent constraints on the formation and evaporation of black holes,” says Strominger, although he acknowledges this work is still at an early stage.

It is just one more setting in which Noether’s theorem looms large, and the list of examples keeps growing. “The relationship between symmetries and conservation laws is a never-ending story,” says Strominger. “One hundred years later, Noether’s theorem keeps finding more and more applications.”

While no one knows what will come next, the incredible power, and longevity, of Emmy Noether’s theorem is undeniable. **D**

Steve Nadis, a contributing editor to *Discover and Astronomy*, is co-author of *From the Great Wall to the Great Collider*.



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